

Bharat Raj Dhaka

bharatrajdhaka5@gmail.com | +91 9256506877 | Portfolio | GitHub | LinkedIn

Summary

Electronics Instrumentation & Control Engineering undergraduate passionate about Microelectronics and VLSI Design, with practical experience in semiconductor device modeling, RTL design, RTL-to-GDSII physical design, custom IC design, functional verification, FPGA implementation, and embedded systems. Proficient with Synopsys and Cadence EDA tools, with a strong interest in Physical Design, Verification, Digital IC Design, and Semiconductor Technology.

Education

Engineering College Bikaner

Bachelor of Technology in **Electronic Instrumentation and Control Engineering** | 3rd Sem | CGPA: 7.52/10 | [Result](#)

Bikaner, Rajasthan

Expected 2028

Shri Bhagirath Memo. Pub. S S Sr Sec School

Class XII | Percentage: 81.60% | [Result](#)

Tidiyasar, Churu (Raj.)

2023

Swami Vivekanand Sr Sec School

Class X | Percentage: 89.00%

Gogasar, Churu (Raj.)

2021

SKILLS & TECHNOLOGIES

Embedded Systems & Firmware Development

- **Microcontrollers & Platforms:** Raspberry Pi, ESP32, ESP8266, Arduino, STM32, Renesas Quickconnect Beginners kit 2.0, FPGA Board (ZedBoard & PYNQ Z2).
- **Programming & Frameworks:** C/C++, Python, Flutter (Dart), HTML, Verilog HDL, System Verilog, Android Studio, VS Code, Arduino IDE, STM32 Cube IDE, Proteus.
- **Communication Protocols:** UART, SPI, SAI, I²C, I2S, Wi-Fi, Bluetooth, GSM, MQTT, ZigBee.

Hardware Design & Prototyping

- **PCB Design Tools:** Proficient in **KiCad** and **EasyEDA** for schematic capture and PCB layout.
- **Prototyping & Fabrication:** Hands-on experience with **PCB fabrication techniques**, including etching and board development processes.
- **Testing & Debugging:** Skilled in hardware diagnostics using **Digital Storage Oscilloscope**, **multimeter**, and signal probing.
- **Fault Analysis:** Strong understanding of **fault analysis** and **circuit behavior diagnostics** for complex electronic systems.

VLSI Tools & Technologies

- **HDL & Digital Design:** Proficient in **Verilog HDL**, **digital electronics**, and **SystemVerilog**.
- **RTL Design & Verification:** RTL Design, Simulation, Functional Verification, Waveform Debugging, Linting, Synthesis, DFT Fundamentals.
- **Physical Design:** RTL-to-GDSII Flow, Floorplanning, Placement, Clock Tree Synthesis (CTS), Routing, Timing Analysis & Optimization.
- **Semiconductor Device Engineering:** Device Modeling, Process Simulation, Device Simulation, Electrical Characterization, Parameter Extraction.
- **EDA Tools:** Hands-on experience with **Synopsys VCS** (RTL simulation), **Verdi** (Waveform Debugging & Analysis), **SpyGlass** (RTL Linting & Quality Checks), **Design Compiler** (RTL Synthesis), **TestMax**

ATPG (DFT & Automatic Test Pattern Generation), **Synopsys Sentaurus TCAD** (Semiconductor Process & Device Simulation), **Mystic** (Electrical Characterization & Parameter Extraction), **Cadence Virtuoso** (Custom IC Design & Analog Layout), **Cadence Genus** (Logic Synthesis), **Cadence Innovus** (Physical Design (Floorplanning, Placement, CTS, Routing)), **Cadence Xcelium** (Functional Simulation & Verification), **Calibre** (DRC, LVS & Physical Verification) and **Xilinx Vivado** (FPGA Synthesis, Implementation & Bitstream Generation).

- **FPGA Platforms:** Practical experience in **FPGA-based design implementation and validation**, including synthesis, bitstream generation, and on-hardware testing.

Experience

STEM Educator & Aeromodeller

learning by doing education / Full-time

Hybrid

Aug 2024 – Present

- Conducting hands-on STEM sessions and aeromodelling activities; mentoring learners through build-based learning.
- Supporting demonstrations, practical prototyping, and iterative improvements during sessions.

Undergraduate Student

Innovation Centre & Product Design Lab, Dept. of E&CE / Full-time

Bikaner, Rajasthan, India (On-site)

Jul 2024 – Present

- Working on VLSI Design, embedded systems and electronic circuit design: prototyping, sensor interfacing, PCB design, and system-level debugging.
- Collaborating on innovation-driven projects in automation, instrumentation, IoT, and signal processing.
- Supporting **VLSI lab** activities, including RTL design/simulation support (**Verilog**); helping with demos, hands-on sessions, and iterative improvements.

Microelectronics & VLSI Design Summer Intern **Certificate** **Surat, Gujarat, India (On-site)**

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat / Internship

June 2026 – July 2026

- Gained hands-on experience in the complete semiconductor and VLSI design flow using industry-standard Synopsys and Cadence EDA tools.
- Performed semiconductor device modeling and simulation using Synopsys Sentaurus TCAD, including device structure design, process simulation, device simulation, and electrical parameter extraction with Mystic.
- Worked on the RTL-to-GDSII Physical Design flow using Cadence Genus and Cadence Innovus, covering synthesis, floorplanning, placement, clock tree synthesis, routing, and timing optimization.
- Designed and analyzed custom IC layouts using Cadence Virtuoso and performed physical verification (DRC/LVS) with Calibre.
- Carried out RTL simulation, functional verification, and waveform debugging using Synopsys VCS, Verdi, and Cadence Xcelium.
- Strengthened practical knowledge of semiconductor device engineering, physical design, digital implementation, verification methodologies, and the end-to-end VLSI design flow.

STEM & Aeromodelling Intern **Certificate**

Center for Creative Learning, IIT Gandhinagar (CCL IITGN) / Internship

Bhopal, Madhya Pradesh, India (On-site)

Sep 2024 – Oct 2024

- Assisted in STEM and aeromodelling activities; supported workshop execution and hands-on learning modules.
- Helped improve learning workflows and project demonstrations.

Patent

Educational Robotics Training Kit with Pre-Wired Embedded Components and Multi-Language Programming

Application Number: **202511113337**

Projects

ECB Robo Training Kit (Embedded Systems Trainer Board)

[Github](#)

- Developed a multi-module **embedded training kit** for hands-on learning and demonstrations on sensors, actuators, and interfacing.
- Implemented interfacing and experimentation with sensors including **DHT22**, **LDR**, **Hall-effect sensor**, **IR receiver**, and **Ultrasonic** sensor.
- Integrated outputs and control modules such as **DC motor driver**, **relay switching**, **buzzer**, and **RGB/LED indicators** for real-time feedback.
- Designed a user interface using a **16×2 I²C LCD** and on-board inputs (switches/potentiometers) for parameter tuning and interactive demos.
- Performed end-to-end testing, debugging, and iterative improvements during lab/training sessions to ensure reliable operation of all modules.

Bird Scaring System — IoT (ESP32, MQTT, RTC, Wi-Fi)

[Github](#)

- Developed an ESP32-based embedded IoT system demonstrating real-world communication and autonomous device control.
- Integrated Wi-Fi, MQTT messaging, real-time clock synchronization, and hotspot connectivity.
- Wrote embedded C/C++ firmware for sensor reading, network operations, MQTT handling, and state management.

Robotic Hand System — Bluetooth-Controlled (Arduino Nano, HC-05)

[Github](#)

- Designed and fabricated a Bluetooth-controlled robotic hand system focusing on PCB design and embedded control.
- Implemented wireless control using HC-05 and firmware generating PWM signals for multiple servo motors.
- Covered end-to-end workflow: schematic, PCB layout, fabrication, assembly, and testing.

Solar Light Controller — ESP8266 (PCB, Fabrication & Programming)

[Github](#)

- Designed and programmed an ESP8266-based solar light control system for day/night automation.
- Implemented battery charge management logic and LED-based status indication.
- Built a custom PCB including power routing, sensor inputs, and control outputs.

8-bit ALU (Add, Sub, AND, OR)

[Github](#)

- Designed an 8-bit Arithmetic Logic Unit with 4 Operations (Add, Sub, AND, OR).
- Implemented and validated the design on a **ZedBoard FPGA** (synthesis, bitstream generation, and on-hardware testing).

Certificates

Digital Design with Verilog	Jan–Apr 2025	CERTIFICATE	NPTEL
Electronic Systems Design..	Jul–Oct 2025	CERTIFICATE	NPTEL
VLSI Design Flow: RTL To GDS	Jul–Oct 2025	CERTIFICATE	NPTEL
Microelectronics: Devices to Circuits	Jul–Oct 2025	CERTIFICATE	NPTEL
VLSI Physical Design with Timing Analysis	Jan–Apr 2026	CERTIFICATE	NPTEL
Introduction to Embedded System Design	Jan–Apr 2026	CERTIFICATE	NPTEL